An Agent-based Model to Provide Decision Support for Traffic and Visitor Management in the Kruger National Park: A Prototype

KNP Visitor Management Model: Final Report

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Summary

This report presents the outcome of a joint project between South African National Parks (SANParks) and the Hamburg University of Applied Sciences. The project entails the development of a prototypical agent-based model (ABM) of the traffic system of the Kruger National Park (KNP). The purpose of the ABM is to enable the simulative study and analysis of traffic patterns in the KNP, as well as intervention testing in a low-cost and low-impact virtual environment. This is motivated by ongoing road congestion challenges in the KNP that SANParks is confronted with, as well as the inherent difficulty of rolling out counter-congestion interventions in a complex and adaptive traffic system. In the report, the design of the model is presented in detail. This presentation is followed by an outline of four hypothetical simulation scenarios. For two of these scenarios, simulations were run and first model results are shown. The report concludes with a critical evaluation of the results and the current state of development of the model. Potential improvements and next steps in the development and enhancement process of the model are also listed. Being a prototypical implementation, the model is not finished—thus, one of the goals of this report is to shed light on its current state and illustrate its potential value to SANParks and other stakeholders.

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Acronyms

ABM Agent-Based Model. 1, *Glossary:* ABM

ASC ASCII raster. 14

ASCII American Standard Code for Information Interchange. iii, 14

CSV Comma-Separated Values. 14

FIFO First In First Out. 5

GeoJSON Georeferenced JavaScript Object Notation. 10

GIS Geographic Information System. 10

JSON JavaScript Object Notation. 12

KNP Kruger National Park. 1

MARS Multi-Agent Research and Simulation. 1, *Glossary:* MARS Framework

OSV Open Safari Vehicle. 8

POI Point of Interest. 5, *Glossary:* POI

SANParks South African National Parks. 1

SHP Shapefile. 10

SOH SmartOpenHamburg. 5

UTC Coordinated Universal Time. 10, *Glossary:* UTC

W3C World Wide Web Consortium. 10

WKT Well-Known Text. 15, *Glossary:* WKT

1. Introduction

There are many reasons to aim for an optimal balance between low congestion and a high number of visitors in South Africa's national parks. These range from economic and financial concerns, visitor satisfaction, and sustainability—making this issue relevant from many points of view. This is especially true for the Kruger National Park (KNP): the largest, most famous, and most frequented national park in South Africa. Seasonally high traffic volume that leads to above-average road congestion is anecdotally known to regular visitors of the KNP and empirically evident to South African National Parks (SANParks) and its affiliates. In Figure 1, two photographs of two-way traffic jams in the KNP that were randomly selected online are shown.

Figure 1: Random selection of online photographs of two-way traffic jams in the KNP. Sources: Seery (2015) (left) and Büscher et al. (2022) (right).

Over the years, numerous investigations have been launched due to concerns about road congestion problems in national parks. For example, in 2014, an impact assessment on traffic was performed in preparation for the construction of the Skukuza Safari Lodge (Havenga 2016). Before and since then, research has been focused on addressing congestion problems in various ways. Some examples are an initiative against overcrowding in the KNP (Ferreira et al. 2014), a comparison of the needs of local and international visitors (Morrison-Saunders et al. 2019), and a reconciliation attempt of stakeholder concerns and traffic management needs through focus groups and interviews (Ballantyne et al. 2023).

Traffic systems are complex, and interventions to manage them tend to be costly, uncertain, time-intensive and unpredictable in terms of amortisation, or all of the above. This complicates the direct implementation of large-scale interventions in real-world traffic systems. In this report, a modelling and simulation approach is presented that enables the study of the KNP traffic system and the testing of interventions in a virtual setting. To this end, a prototypical agent-based model (ABM) of the KNP was developed by the Multi-Agent Research and Simulation (MARS) Group in conjunction with SANParks. The rest of the report is structured as follows. In Section 2, the basic concepts of agent-based modelling are introduced. In Section 3, the model is described in detail. In Section 4, some potential scenarios for the model to simulate are outlined. In Section 5, simulation results for two of these scenarios are shown. In Section 6, the results and the current state of the model are evaluated. In Section 7, a brief conclusion finalises the report.

2. Definitions and Concepts

In preparation for the description of the KNP traffic and visitor management model (hereafter mostly referred to as the "model"), some basic definitions and concepts are provided in this section. Specifically, the main ingredients of agent-based modelling are introduced, followed by an overview of the MARS Framework as a technical means to implement and simulate ABMs.

2.1. Agent-based Modelling

An ABM is a computerised simulation model in which a set of agents share a common environment. The environment of an ABM is the spatial extent in which agents are situated and can move. An agent is an autonomous software entity that executes behaviour in its environment to achieve some desirable state (e.g., by accomplishing a set of goals).

In Figure 2, a common visual representation of an ABM is shown. The agent carries out some behaviour in its environment. As a result, the state of the environment might change. The agent can perceive some or all of these changes, which can enable and inform its subsequent behaviours. This modelling approach enables the study of how behaviours of individuals in a system impact the global dynamics of that system.

A simulation of an ABM is produced by performing consecutive iterations (also called simulation steps) of the cycle of behaviour and interaction shown in Figure 2 over some simulation time period. To simulate a finished ABM, one can provide a configuration: a specification of the initial parameter values of the simulation. Given an initial model state, the simulation can run its course and produce a set of results, allowing modellers to infer relationships between the input and the output of the ABM. It is important to note that typical ABMs comprise not just one agent but multiple agents that interact with the environment simultaneously and can also interact with each other. This contributes to the complexity as well as to the expressiveness of agent-based modelling.

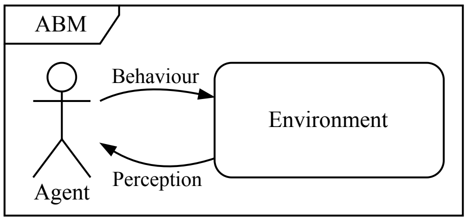

Figure 2: ABM from the point of view of a single agent. The agent interacts with its environment and perceives changes in the environment in an ongoing interaction loop.

For more details on these topics, we recommend the following resources:

- Agent-based modelling: Weiss (2000), Wooldridge (2002), Shoham et al. (2009), and Dorri et al. (2018)
- Complex adaptive systems: Gros (2015) and Sayama (2015)
- Agent-based modelling and geographical data: Castle et al. (2006) and Crooks et al. (2018)
- Agent-based modelling and road traffic management: Bazghandi (2012), Adesina et al. (2017), and Yun et al. (2022)

2.2. The MARS Framework

The MARS Framework is developed by the MARS Group (2023) at the Hamburg University of Applied Sciences. It provides two main functionalities: (i) the implementation of ABMs as executable software and (ii) the simulation of ABMs, producing a set of model outputs.

Environment The environment of a MARS model consists of a set of layers. A layer is a cross-section of the modelled environment that contains like objects. The overlay of individual layers forms a cohesive environment that agents can interact with as a whole. In Figure 3, a schematic example of this is shown for a hypothetical ABM in an urban setting: this exemplary environment consists of a road network (grey), a set of buildings (red), a set of traffic lights (yellow), and a set of parking spaces (green).

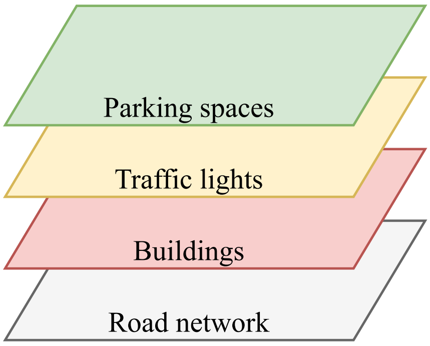

Figure 3: Environment of a hypothetical MARS model in an urban setting, consisting of a set of layers that contain a road network, buildings, traffic lights, and parking spaces.

MARS models can have georeferenced environments. For the hypothetical example shown in Figure 3, this would mean a road network with a georeferenced extent as well as buildings, traffic lights, and parking spaces with georeferenced locations. ABMs with georeferenced environments can model the spatial properties of their real-world counterparts with high accuracy, making for less abstract and more realistic models. In addition, the time period of a simulation of a MARS model can be expressed in standard date and time formats. This enables a close link between the temporal rates at which developments occur in the real world and in the model.

Agent An agent of a MARS model is described by the following:

1. a set of **properties** such as the agent's state, knowledge, memories, goals, etc.,
2. an **initialisation routine** that is executed once at the simulation start, and
3. a **behaviour routine** that is executed during each simulation step.

The initialisation routine ensures that agents start a simulation as specified in the configuration. The behaviour routine enables the in-simulation interaction cycle shown in Figure 2.

3. The KNP Visitor Management Model

For this project, the MARS Framework was used to implement a configurable ABM that can simulate traffic flow in the KNP. To this end, domain knowledge, expertise, and various datasets were provided by SANParks to inform and parametrise the model. In this section of the report, a detailed description of the model is provided. This includes (i) an overview of the main components of the model, (ii) an outline of the data provided by SANParks and how these data are integrated into the model, and (iii) a general guide to configuring the model to suit a range of simulation scenarios.

3.1. Model Description

The main components of the model and the relationships between them are shown in Figure 4. A description of each component and its contents follows.

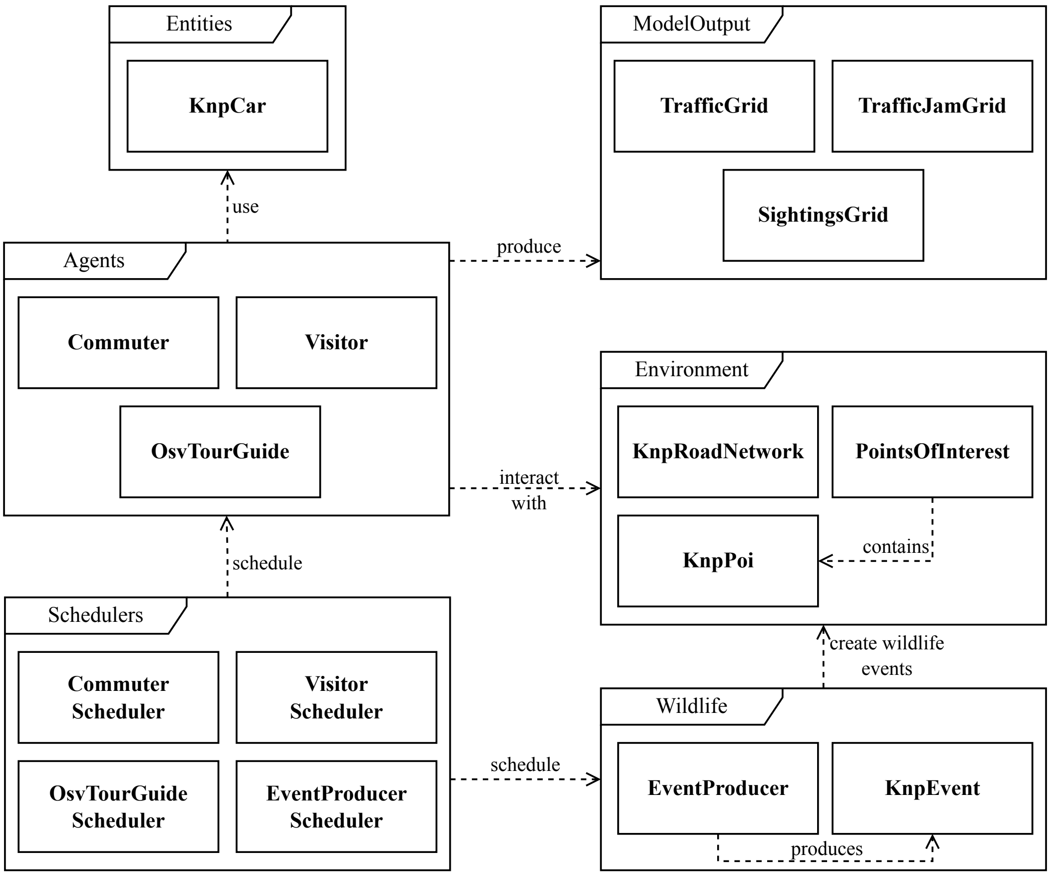

Figure 4: High-level overview of the main components (outer frames) and contents (rectangles inside frames) of the model, along with the relationships between them (arrows).

3.1.1. Environment

The environment of the model consists of two layers: the `KnpRoadNetwork` and the `PointsOfInterest`.

The `KnpRoadNetwork` layer is designed to hold the road network of the KNP, consisting of a set of georeferenced road segments. A road segment is described by a name, length, speed limit, surface type, access restriction, and other properties. The movement of agents through the KNP is limited to the course of the road segments. The `KnpRoadNetwork` layer also provides routing services to agents: computing the shortest path between two points, computing the duration of a given route, etc.

The `PointsOfInterest` layer is designed to hold a set of points of interest (POIs). A POI is a special georeferenced location in the environment (e.g., a KNP gate or a rest camp) that agents can travel from or to. Each POI of the KNP is modelled as a `KnpPoi` feature, consisting of a georeferenced position, name, type, access restriction, and other properties. Agents can query the `PointsOfInterest` layer for information about `KnpPoi` features: finding the nearest feature of a given type, finding all features in a given region, etc. This information can help agents plan their trips on the KNP road network.

Note: Since the focus of this model is on traffic flow on the KNP road network, the internal structure of POIs is not represented in the model.

3.1.2. Wildlife

In the model, wildlife occurrences are modelled as georeferenced point events that can appear temporarily on the roadside. A wildlife event is represented as a `KnpEvent` with a location and duration. The `EventProducer` is responsible for the creation and removal of `KnpEvents` over time. Agents can perceive these events with some probability and, if identified, react accordingly.

3.1.3. Entities

The `KnpCar` is a passive entity in the model. It represents a standard passenger vehicle that can be used by agents to move on the KNP road network. Its movements along the road network are modelled according to basic laws of physics; for example, the processes of accelerating and decelerating happen over calculated time periods to model realistic traffic flow. The `KnpCar` also follows the First In First Out (FIFO) principle when the right-of-way sequence at intersections needs to be determined.

Note: The base implementation of the `KnpCar` entity was developed as part of the SmartOpenHamburg (SOH) project. In this project, the MARS Group implemented multimodal traffic models of the city of Hamburg, Germany. For more information, see Clemen et al. (2021) and Lenfers et al. (2021).

3.1.4. Agents

The model features Commuter, Visitor, and OsvTourGuide agents. The behaviour routine of each agent is described below.

💡 **Note:** In the model, agent behaviours and decisions are stochastic, meaning that they follow some probability distribution. Therefore, any two agents that are configured the same way will not necessarily produce the same behaviour during a simulation. This is done to provide a decision space to the agents that is representative of the variability of decisions made by humans in the real world.

Commuter A Commuter agent represents a person who lives outside the KNP and commutes into the KNP daily for work. The daily behaviour routine of a Commuter agent is shown in Figure 5.

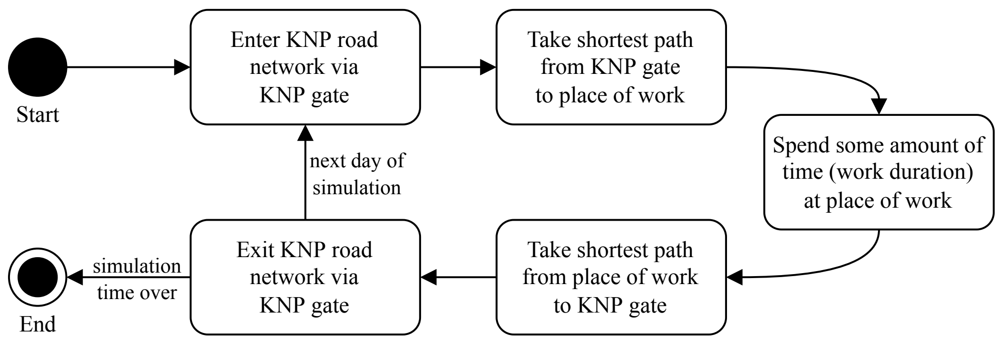

Figure 5: Daily behaviour routine of a Commuter agent.

A Commuter agent starts its behaviour routine at a KnpPoi of the type KNP Gate. From there, it takes the shortest path to its place of work (e.g., a KnpPoi of the type rest camp that is not too far away from the KNP Gate). The agent spends some amount of time (work duration) at its place of work, before taking the shortest path back to a KNP gate (e.g., the one from which it entered) to exit the KNP. This daily routine can be repeated for the duration of the simulation.

Visitor A `Visitor` agent represents a person who visits the KNP for some time and explores the KNP for wildlife. The daily behaviour routine of a `Visitor` agent is shown in Figure 6.

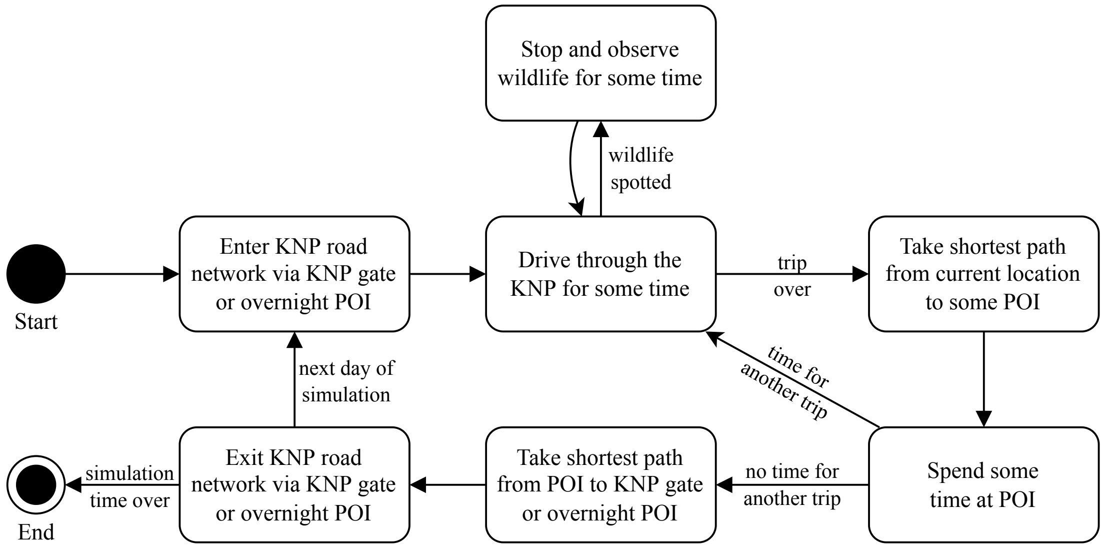

Figure 6: Daily behaviour routine of a single-day or an overnight `Visitor` agent.

A `Visitor` agent starts its behaviour routine at a `KnpPoi` of the type KNP Gate or at a POI at which one can spend the night (referred to as an overnight POI in Figure 6). This corresponds to a single-day `Visitor` or an overnight `Visitor`, respectively. From its starting position, the agent begins to explore the KNP in search of wildlife by driving randomly on the road network. When it spots wildlife, the `Visitor` agent brakes and observes it for some time before continuing to explore randomly. After some time, the agent stops random exploration and begins to take the shortest path to some nearby POI. This corresponds to a person visiting, for example, a rest camp or picnic spot for a bathroom break and/or to eat. The cycle of (i) exploring randomly and stopping for spotted wildlife for some time, (ii) taking the shortest path to some POI, and (iii) taking a break at that POI continues for as long as there is time during the current day. Finally, the `Visitor` exits the KNP road network via a KNP Gate or an overnight POI.

OsvTourGuide An `OsvTourGuide` agent represents a person who drives an open safari vehicle (OSV) with visitors through the KNP. The daily behaviour routine of an internal `Osv-TourGuide` (i.e., one that is based at a POI that can host OSVs) is shown in Figure 7.[1]

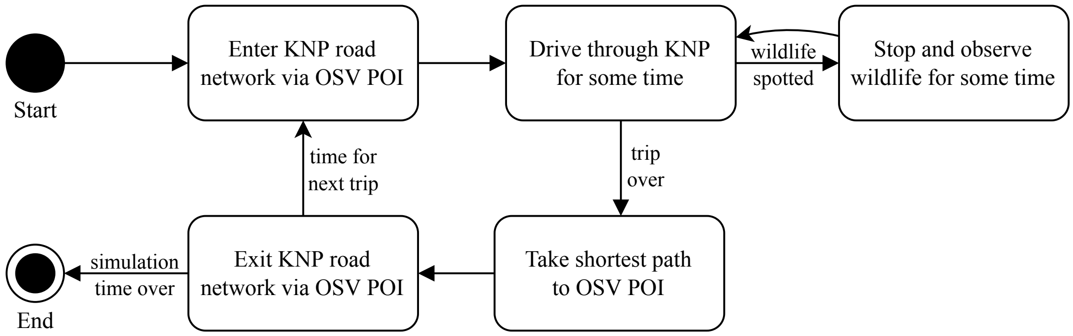

Figure 7: Daily behaviour routine of an `OsvTourGuide` agent that is based at an OSV POI.

An `OsvTourGuide` agent starts its behaviour routine at a `KnpPoi` of the type KNP Gate or at a POI that can host OSVs (referred to as an OSV POI in Figure 7). From there, the agent begins to explore the KNP in search of wildlife by driving randomly on the road network. The cycle of (i) exploring randomly and stopping for spotted wildlife, (ii) taking the shortest path to the rest camp, and (iii) waiting until it is time for the next trip is similar to the cycle of the `Visitor` agent (see Figure 6).

3.1.5. Model Output

During a simulation of the model, data are generated and collected as a result of decisions and behaviours of the agents. These data are collected by the model output component for later analysis outside the context of the MARS Framework. One of the output formats of the model is georeferenced raster data: grids that are superimposed onto the KNP road network to track different information during a simulation. The three layer types shown in the model output component of Figure 4 serve the following functions:

- `TrafficGrid`: tracks the movement density on the KNP road network
- `TrafficJamGrid`: tracks the location and duration of traffic jams on the KNP road network
- `SightingsGrid`: tracks the location and duration of wildlife sightings on the KNP road network

[1] The daily behaviour routine of an external `OsvTourGuide` (i.e., one that enters the KNP through a KNP Gate daily) is similar to the daily behaviour routine of a single-day `Visitor` agent (see Figure 6). The main difference is that an external `OsvTourGuide` does not enter or exit the KNP road network via an overnight POI but rather via a KNP Gate.

These outputs can be used to produce heatmaps for qualitative analysis of traffic density, traffic jam intensity, and wildlife sighting durations across the KNP road network, respectively.

3.1.6. Schedulers

The scheduling component of the model contains a set of schedulers that enables the addition of new agents to an ongoing simulation at specific locations and time points. `Commuter` agents, `Visitor` agents, and `OsvTourGuide` agents can be scheduled via the `CommuterScheduler`, the `VisitorScheduler`, and the `OsvTourGuideScheduler`, respectively. In addition, `EventProducers` can be scheduled to spawn wildlife events across the model environment via the `EventProducerScheduler`. The scheduling patterns of a simulation can be configured by specifying spawn periods. A spawn period is a time period during a simulation during which spawn events occur. A spawn event is an event during a spawn period in which new agents join an ongoing simulation.

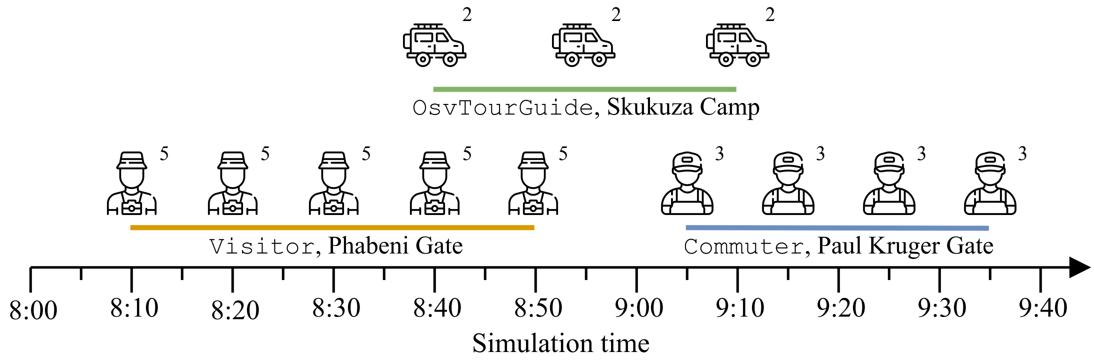

Figure 8: Three partially overlapping spawn periods (orange, green, and blue) with different durations that occur in different time periods of an ongoing simulation. The spawn periods contain spawn events that add (from left to right) `Visitor` agents, `OsvTourGuide` agents, and `Commuter` agents to the simulation at different amounts (numbers at the top right of the icons), frequencies, and locations.

In Figure 8, an illustration of different spawn periods over the time period of a simulation is shown. This example features three spawn periods:

- The orange spawn period ranges from 8:10 to 8:50 and contains five spawn events, each spawning five `Visitor` agents at the Phabeni Gate. Such a spawn period can be configured via the `VisitorScheduler`.
- The green spawn period ranges from 8:40 to 9:10 and contains three spawn events, each spawning two `OsvTourGuide` agents at the Skukuza Rest Camp. Such a spawn period can be configured via the `OsvTourGuideScheduler`.
- The blue spawn period ranges from 9:05 to 9:35 and contains four spawn events, each spawning three `Commuter` agents at the Paul Kruger Gate. Such a spawn period can be configured via the `CommuterScheduler`.

Table 1: Parameters of a spawn period, including their data type and whether they are required or optional for defining a valid spawn period.

Name	Type	Description	Required
`startTime`	UTC Datetime	Time point at which the spawn period begins.	✔
`endTime`	UTC Datetime	Time point at which the spawn period ends.	✔
`spawningInter-valInMinutes`	Positive integer	Number of minutes between consecutive spawn events in the spawn period.	✔
`spawningAmount`	Positive integer	Number of agents to be spawned per spawn event.	✔
`sourceName`	String	Name of the POI at which agents spawn.	✔
`sourceType`	String	Type of the POI specified in `sourceName`.	✔
`targetName`	String	Name of the POI to which agents travel first.	✖
`targetType`	String	Type of the POI specified in `targetName`.	✖

The general parameters with which a spawn period can be configured are listed in Table 1. The `startTime` and `endTime` of a spawn period can be expressed in the Coordinated Universal Time (UTC) datetime format.[2] To set the spawn location to a specific POI, the parameters `sourceName` and `sourceType` can be used (for example, `sourceName = Phabeni` and `sourceType = KNP Gate` to spawn agents at the Phabeni Gate).

3.2. Data Integration

By offering the functionality outlined in Section 2.2, the MARS Framework enables the design of ABMs with georeferenced environments that can be used to simulate behaviour over relatable time periods. However, to bring the components of the KNP visitor and traffic management model outlined in Section 3.1 to life, the right real-world datasets are needed. For this project, SANParks provided a number of datasets which were integrated into the model to align it with its real-world counterpart. In this section, each dataset and the way it is integrated into or associated with the model is described.

3.2.1. Georeferenced Environment Data

As mentioned in Section 3.1.1, the environment of the model consists of two layers: (i) the `KnpRoadNetwork` layer, which is designed to hold the KNP road network, and (ii) the `PointsOfInterest` layer, which is designed to hold the POIs of the KNP.

To inform the contents of the `KnpRoadNetwork`, SANParks provided a digital vector representation of the KNP road network as a Shapefile (SHP). The contents of the SHP file were transferred to a Georeferenced JavaScript Object Notation (GeoJSON) file to (i) enable easier integration into the MARS Framework and (ii) enable easier visualisation of the file contents with open-source Geographic Information System (GIS) tools.[3]

[2]For more information on UTC datetime formats, see this World Wide Web Consortium (W3C) note.
[3]For more information on GeoJSON, see this Wikipedia entry.

SANParks also provided a set of SHP files containing the point locations of different types of POIs in the KNP. These data are used to create `KnpPoi` objects that are held by the `PointsOfInterest` layer. In Appendix A, a full listing of the 30 POI types that are available in the model as a result of this dataset is included. In Appendix B, some common publicly accessible POIs are listed.

When a simulation of the model is started, the KNP road network and the POIs from the respective SANParks datasets are automatically integrated into the model. During the data integration process, the MARS Framework creates a content-equivalent in-model representation of the KNP road network and the POIs that is held by the `KnpRoadNetwork` layer and the `PointsOfInterest` layer, respectively. In Figure 9, a visual representation of the resulting model environment is shown, including the road segments that make up the KNP road network (black lines) and an exemplary subset of six POI types (coloured points).

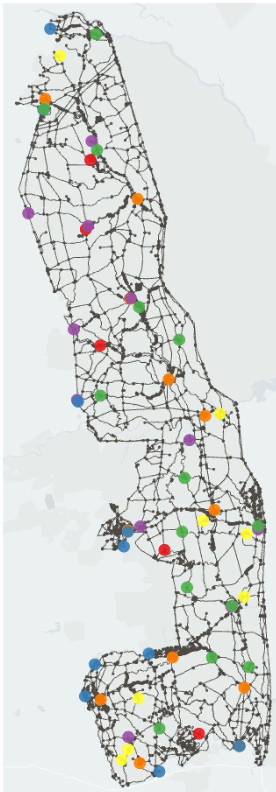

Figure 9: Visual representation of the environment of the model, consisting of the KNP road network (black lines) and a subset of available POIs containing six POI types: Bushveld camp (red), KNP Gate (blue), picnic spot (green), ranger camp (purple), rest camp (orange), and trails camp (yellow).

3.2.2. Rest Camp Capacity and Occupancy

SANParks provided a dataset containing monthly capacity and occupancy counts for various KNP rest camps. The dataset ranges from April 2018 through March 2020. These data are not explicitly integrated into the model; but they can be used to inform the number of agents to be spawned at each of these POIs via the schedulers (see Section 3.1.6) in long-term simulations.

3.2.3. Gate Quotas and Entries

Analogous to the capacity and occupancy data for KNP rest camps (see Section 3.2.2), SANParks provided quotas and entry counts for the KNP gates. In Table 2, the daily quota of each KNP gate is listed. The entry count dataset is aggregated hourly and grouped into weeks ranging from April 2018 through March 2020. Like the rest camp datasets, these data are not directly integrated into the model, but can be used to inform the number of agents to be spawned at each KNP gate on an hourly basis.

Table 2: Daily gate quotas.

KNP Gate	Quota
Crocodile Bridge	1100
Giriyondo	275
Malelane	800
Numbi	800
Orpen	700
Pafuri	600
Paul Kruger	800
Phabeni	800
Phalaborwa	800
Punda Maria	700

3.2.4. OSV Permits

SANParks provided daily OSV permit data per KNP gate. These data include the number of external OSVs that entered the KNP through each KNP gate between April 2018 and March 2020. Like the rest camp data (see Section 3.2.2) and the KNP gate data (see Section 3.2.3), these data are currently not integrated into the model, but can be used to inform the number of `OsvTourGuide` agent to be spawned at each KNP gate per simulation day.

3.3. Model Configuration

Based on the model description (see Section 3.1) and the data provided by SANParks to inform the model (see Section 3.2), the configuration of the model to run simulations can be described. In this section, an outline of the configuration options is provided to serve as a guide to configuring the model for different scenarios.

The main configuration file of the model is a JavaScript Object Notation (JSON) file named `config.json`. In Figure 10, a visual representation of the hierarchical structure of the JSON file is shown. The file contains configuration blocks for global simulation parameters, layer parameters, entity parameters, and agent parameters. Each block may contain references to auxiliary configuration files that are stored in a directory named `resources`. Each block of `config.json` and the purpose of the auxiliary files is described below in turn.

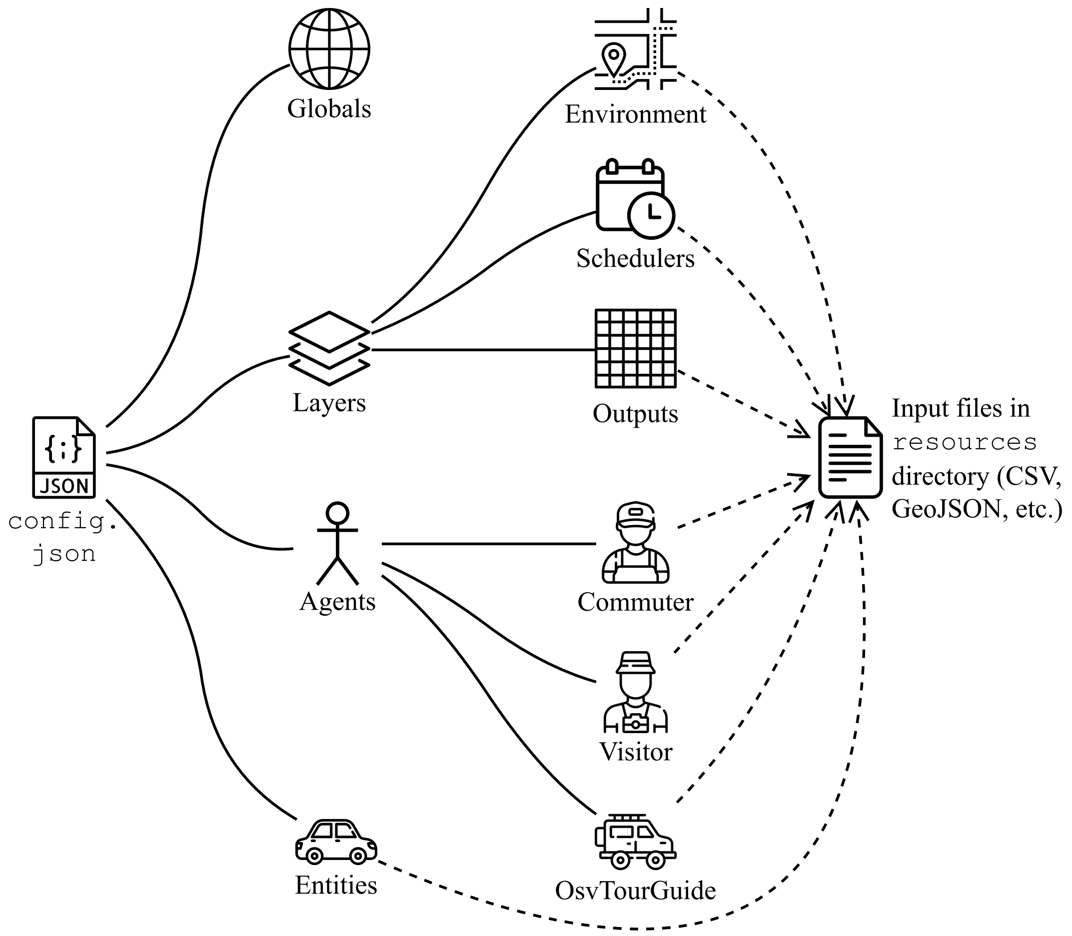

Figure 10: Visual representation of the hierarchical structure of the JSON configuration file (`config.json`) of the model.

3.3.1. Globals

In this block of `config.json`, global properties of a simulation can be configured. The most important parameters are listed in Table 3.

Table 3: Main global configuration parameters of a simulation of the model. "n/a" is used to indicate that the parameter has no default value.)

Name	Type	Default value	Description
`deltaT`	Positive integer	1	The magnitude of one simulation step.
`deltaTUnit`	String	"seconds"	The unit of one simulation step.
`startTime`	UTC Datetime	n/a	The start time of the simulation.
`endTime`	UTC Datetime	n/a	The end time of the simulation.

⚠ **Warning:** Do not change the values of `deltaT` or `deltaTUnit`. The simulation step of the model needs to be scaled to seconds to maintain realistic movement physics of the `KnpCar` entities (see Section 3.1.3 for more details).

3.3.2. Layers

In this block of `config.json`, layer properties of a simulation can be configured. This includes the environment component (see Section 3.1.1), the model output component (see Section 3.1.5), and the scheduling component (see Section 3.1.6). A description of the configuration of each component follows.

Environment To create the environment, the `KnpRoadNetwork` layer and the `PointsOfInterest` layer need to be populated with data. The `KnpRoadNetwork` layer takes a GeoJSON file containing the KNP road network. The default file is named `roads_all_2019_inferred.geojson` and is located in the `resources` directory. The `PointsOfInterest` layer also takes a GeoJSON file. The default file is named `pois_inferred.geojson` and is located in the `resources` directory.

Model output Each of the three output raster layers `TrafficGrid`, `TrafficJamGrid`, and `SightingsGrid` takes an ASCII raster (ASC) file. These files contain a grid that is superimposed onto the KNP road network. This enables the tracking of cumulative data related to traffic density, traffic jam intensity, and wildlife sighting duration during the simulation. The default file for the three layers is named `knp_raster_1111m.asc` and is located in the `resources` directory. The file contents encode a grid of cells, each of which has an area of approximately 1 km^2. For a more fine-grained grid with cells that have an area of approximately 100 m^2, the file `knp_raster_111m.asc` is also provided in the `resources` directory.[4]

At the end of a simulation, the raster data of each of the three layers are written to a GeoJSON file that can be used to visualise the cumulative data collected during the simulation.

Schedulers Each of the four schedulers (the `CommuterScheduler`, the `VisitorScheduler`, the `OsvTourGuideScheduler`, and the `EventProducerScheduler`) takes a Comma-Separated Values (CSV) file that contains the parameters of the spawn periods for the scheduler. The CSV files are located in the `resources` directory. Each row of a CSV file specifies a spawn period. The values in a row should be separated by commas.[5]

[4] For more information on ASC files in the MARS Framework, see the MARS documentation on ASC files.
[5] For more information on CSV files in the MARS Framework, see the MARS documentation on CSV files.

Table 4: Exemplary scheduler CSV file containing a header (first row) and configurations of four spawn periods (second through fifth row). The contents of the header row are rotated for readability. The leftmost column serves only to index the rows and would not be part of a corresponding CSV file.

Spawn period	`startTime`	`endTime`	`spawning-IntervalInMinutes`	`spawning-Amount`	`sourceName`	`sourceType`	`targetName`	`targetType`
1	8:00	9:00	30	5	Phabeni	KNP Gate		
2	8:30	10:30	5	2	Skukuza	Rest camp	Phabeni	KNP Gate
3	9:20	14:30	20	8	Berg-en-Dal	Rest camp		
4	12:00	16:00	15	8	Paul Kruger	KNP Gate		

In Table 4, an example of a scheduler CSV file with four spawn periods is shown. In this example, some optional values are left blank. For more details on the attributes in the header row, see Table 1. For a comprehensive list of POI types, see Appendix A. For a list of some POIs with public access, see Appendix B.

Note: The scheduler CSV files can also be used to set agent-specific parameter values per spawn period. To do so, an header entry with the parameter name needs to be added to the header of the CSV file, and the new column needs to be populated with values for the spawn periods defined in the CSV file.[6]

Note: An optional alternative to using the parameters `sourceName` and `sourceType` or `targetName` and `targetType` is to specify a `sourceGeometry` and `targetGeometry`, respectively. This might be useful for simulation scenarios in which agents are not required to be spawned at a specific POI or travel to a specific POI. To parametrise locations via the `sourceGeometry` or `targetGeometry` parameters, a column with the respective name needs to be added to the scheduler CSV file. The values in the new column need to be Well-Known Text (WKT) Strings.[7] For example, the WKT String listed in Figure 11 contains a set of coordinates that correspond to the dark-coloured rectangle shown in Figure 12. Setting the parameter `sourceGeometry` to this WKT String would cause the initial position of agents that are spawned during the spawn period to be on any road segment that lies within this geometry. The same mechanism applies to `targetGeometry` with respect to the first trip destination of the agents. When using `sourceGeometry` or `targetGeometry`, the parameters `sourceName` and `sourcetype` or `targetName` and `targetType`, respectively, can be left blank.

[6]For more information on configuring schedulers, see the MARS documentation on scheduler parameters.

[7]For more information on WKT vector geometry formats and syntax, see this Wikipedia entry.

```
POLYGON (
   (31.487209076653926, -24.972101229643584,
   31.487209076653926, -25.019197226970533,
   31.624840397979995, -25.019197226970533,
   31.624840397979995, -24.972101229643584,
   31.487209076653926, -24.972101229643584)
)
```

Figure 11: Exemplary WKT String encoding a rectangle that covers a small section of the KNP around the Skukuza Rest Camp. See Figure 12 for a visual representation of this WKT String.

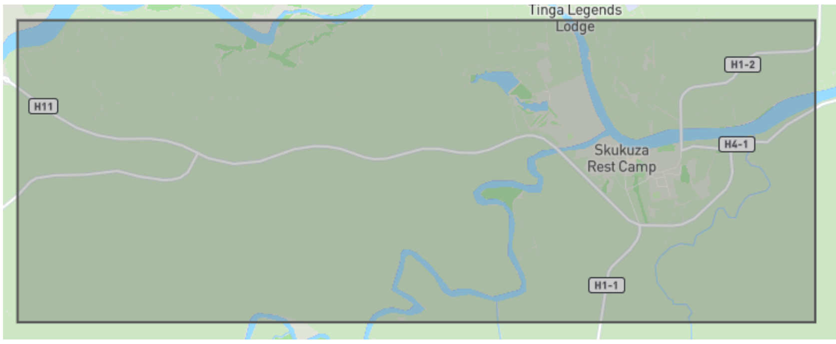

Figure 12: Visual representation of a WKT String (dark-coloured rectangle) that covers a small section of the KNP around the Skukuza Rest Camp. See Figure 11 for an encoding of this WKT String. Generated with geojson.io.

3.3.3. Entities

The entity `KnpCar` requires a CSV file with some basic dimensional parameters. The parameter values are used to perform physics calculations during the movement of the entity. For an overview of the `KnpCar` entity, see Section 3.1.3.

3.3.4. Agents

The agents `Commuter`, `Visitor`, and `OsvTourGuide` each have a separate section in `config.json`. These sections can be used to set parameter values for all agents.[8] For agent parametrisation at the spawn period level, the schedulers of the respective agent can be used. For an overview of the `Commuter`, `Visitor`, and `OsvTourGuide` agent, see Section 3.1.4.

Commuter The `Commuter` agent has a parameter `workDuration`. The value of this parameter specifies the amount of time (in minutes) that an agent spends at its place of work. For an overview of the daily behaviour routine of the `Commuter` agent, see Figure 5.

[8]For more information on setting such parameters, see the MARS documentation on individual mappings.

4. Scenarios

In Section 3, the model's current state of development and the configuration options for simulations were explained in detail. In this section, a description of the design, configuration, and execution of two simulation scenarios is provided. An attempt has been made to base the scenarios on potential real-world traffic management interventions in the KNP. In addition, two more scenarios that were not simulated for this report, but that can be investigating with the model, are outlined as well.

The global configuration parameters of all performed simulations were set to the values that are listed in Table 5. Each simulation is configured to proceed in seconds and have a duration of six hours (for an overview of global configuration parameters, see Section 3.3.1). The ASC file `knp_raster_1111m.asc` was used for all output grids (for an overview of the model output grids, see Section 3.1.5).

Table 5: Global configuration parameter values for the simulation scenarios, specifying that the simulation proceeds in single seconds and lasts for six hours.

Name	Value
`deltaT`	1
`deltaTUnit`	"seconds"
`startTime`	`2023-06-15T06:00:00`
`endTime`	`2023-06-15T12:00:00`

⚠ **Disclaimer:** The following scenarios and simulation experiments were designed and carried out by the developers of the model and not by domain experts. The main goal of the scenarios is to showcase the model's configurability, functionality, versatility, and potential value to stakeholders. However, parts of the experimental setup might be perceived as unrealistic or not meaningful. The design of meaningful simulation scenarios is best done by domain experts who are more familiar with the KNP and the management needs of SANParks.

4.1. Scenario 1: `Visitor` and `OsvTourGuide` Agent Counts

Every day, many individuals with different motivations and behaviour patterns travel through the KNP. Scenario 1 (S1) aims to address the following question:

How does a change in the number of private vehicles and OSVs in the KNP affect traffic?

This question is motivated by the observation that, on a typical day, there are more personal vehicles than OSVs in the KNP. Therefore, the greatest potential for reducing the total number of vehicles in the KNP is to reduce the number of personal vehicles permitted into the KNP per day. However, doing this also results in fewer people being able to visit the KNP per day. To compensate for this, the number of OSVs per day can be increased. This could be done by, for example, increasing the number of OSV permits (see Section 3.2.4). Since an OSV can seat more people than a standard personal vehicle, the amount of the increase in OSVs is expected to be less than the amount of the decrease of personal vehicles. Therefore, this approach should result in fewer vehicles on the road and, by extension, less traffic.[9]

The model can be used to tackle the question by running simulations that differ in the number of `Visitor` agents and `OsvTourGuide` agents. Comparing the results of these simulations might provide insights into changes in traffic patterns and density. To enable the simulations, two different configurations of the `VisitorScheduler` and `OsvTourGuideScheduler` need to be defined. In Appendix C, the configuration parameters of the `VisitorScheduler` and the `OsvTourGuideScheduler` for Configuration 1 (S1C1) and Configuration 2 (S1C2) are listed. All parameters other than those pertaining to the spawn periods of `Visitor` agents and `OsvTourGuide` agents are kept constant in both simulations. In S1C1, a total of 50 `Visitor` + 12 `OsvTourGuide` = 62 agents are expected. In S1C2, a total of 16 `Visitor` + 28 `OsvTourGuide` = 44 agents are expected.

4.2. Scenario 2: Comparing KNP Gate Quotas

Scenario 2 (S2) aims to address the following question:

How does a change in KNP gate quotas affect traffic?

This question is motivated by the observation that each KNP gate has a daily quota (see Table 2). The daily quota of a KNP gate can be changed to adjust the maximum number of vehicles that can enter the KNP through that gate. If it is empirically found that a certain region of the KNP is especially prone to road congestion, then the daily quotas of nearby KNP gates can be reduced to limit the number of vehicles that can enter through these gates and directly into this region of the KNP. This might result in an overall decrease of traffic, or at least a more even distribution of traffic across the KNP.

The model can be used to tackle the question by running simulations that differ in the number of agents that are spawned at each KNP gate. Comparing the results of these simulations might provide insights into changes in traffic patterns and density. Again, two different configurations are needed. In Appendix C, the configuration parameters for Configuration 1 (S2C1) and Configuration 2 (S2C2) are listed. Any parameters that are not mentioned are kept constant in both simulations. For S2C1, the entry numbers per KNP gate were derived from the real-world KNP gate quotas (see Table 2) and scaled down by a factor of 10. Therefore, in S2C1, 110 agents are expected to enter from the Crocodile Bridge Gate and 80 agents are expected to enter from Malelane Gate and Numbi gate each. In S2C2, these numbers were halved.

[9]This line of reasoning ignores the personal preference that visitors of the KNP might have towards driving in their personal vehicle or in an OSV, as such personal preferences do not factor into the question at hand.

4.3. Scenario 3: Limiting Travel by Road Access

Scenario 3 (S3) aims to address the following question:

How does the closure of certain roads affect traffic?

Traffic flow in the KNP is likely to vary depending on which roads are accessible to which kinds of travellers. The road access levels in the KNP road network dataset provided by SANParks (see Section 3.2.1) are “public”, “private”, and “staff”. Another feature that distinguishes road segments from one another is their surface type: “graded”, “gravel”, “tar”, etc. In early 2023, parts of the KNP were hit by a flood (SANParks 2023). As a result, certain roads segments had to be closed due to safety concerns.

The model can be used to assess the potential impact of such interventions on traffic in the KNP. For this, the model’s configurability can be extended to accept parameters that specify the types of roads that are generally accessible. Alternatively, a more fine-grained configurability might enable users of the model to specify which agents are allowed to access which types of road segments. For example, a simulation could be configured such that all agent can use tar roads only. Given such a configurability, different simulations can be carried out with different access specifications to study the effects on traffic density and traffic jam duration.

💡 **Note:** As mentioned in the preamble, this scenario was not simulated for this report.

4.4. Scenario 4: Adding or Removing Road Segments

Scenario 4 (S4) aims to address the following question:

How does the addition and/or removal of road segments and/or POIs affect traffic?

This question might be motivated by empirical observations that certain parts of the KNP are more prone to traffic than others. As a result, SANParks might consider modifying the KNP road network and/or the distribution of POI in those parts of the KNP in an attempt to improve traffic conditions. For example, the construction of new road segments would make the road network as a whole more dense, which in turn might cause traffic to flow more smoothly.

The model can be used to assess the potential impact of such interventions on traffic in the KNP. To this end, a modified copy of the digital representation of the KNP road network (see Section 3.2.1) or the POIs can be generated. Then, different simulations can be run with the different georeferenced datasets to study any differences in traffic due to the modifications in the environment of the model.

💡 **Note:** This scenario requires the use of GIS tools to generate the new georeferenced datasets.

💡 **Note:** As mentioned in the preamble, this scenario was not simulated for this report.

5. Results

In this section, the results of the first two simulation scenarios outlined in Section 4 are presented. All figures included in this section were generated with kepler.gl.

5.1. Scenario 1

In the following figures, the `TrafficGrid` layers (Figure 13 and Figure 14, respectively), the `TrafficJamGrid` layers (Figure 15 and Figure 16, respectively), and the `SightingsGrid` layers (Figure 17 and Figure 18, respectively) that resulted from S1C1 and S1C2 are presented side by side for comparison. In general, the darker the colour of a square, the more activity (i.e., traffic density, traffic jam duration, or wildlife sighting duration) occurred on the underlying road segment.

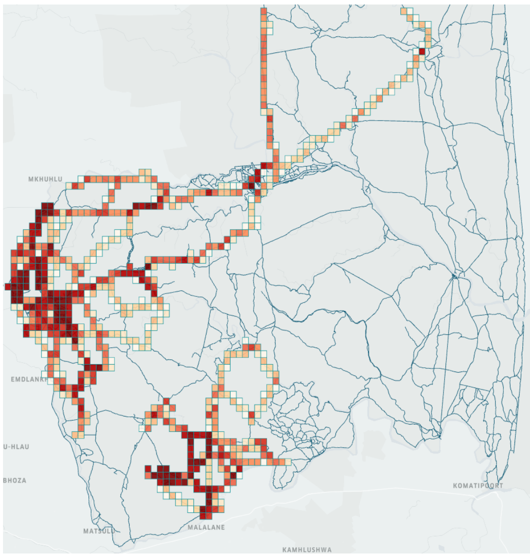

Figure 13: S1C1: Traffic density.

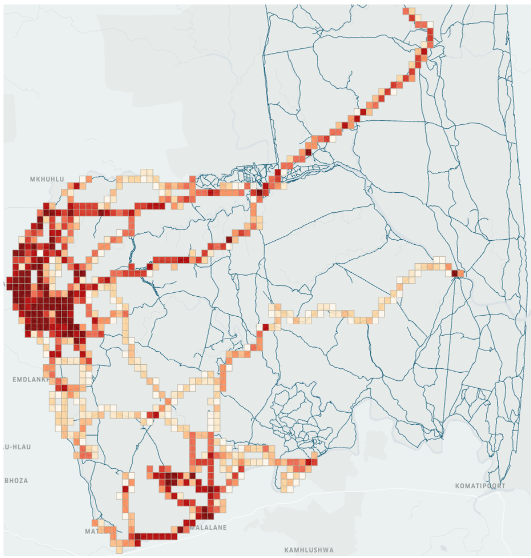

Figure 14: S1C2: Traffic density.

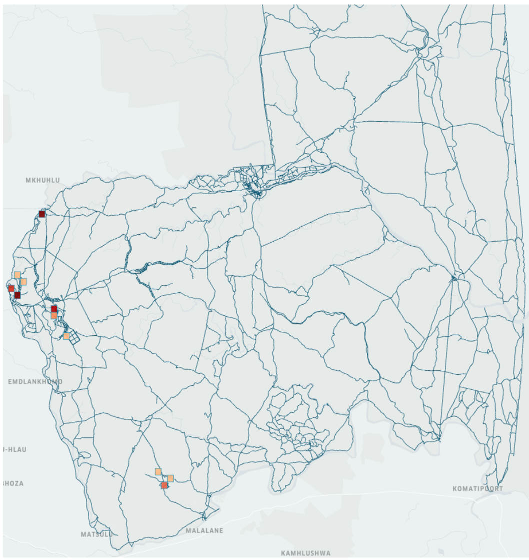

Figure 15: S1C1: Traffic jam locations and densities.

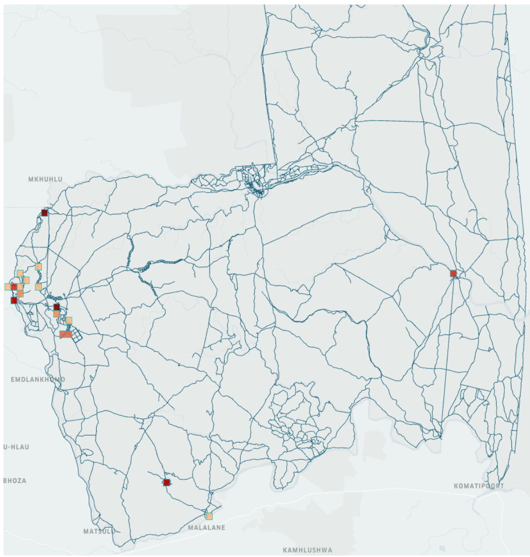

Figure 16: S1C2: Traffic jam locations and densities.

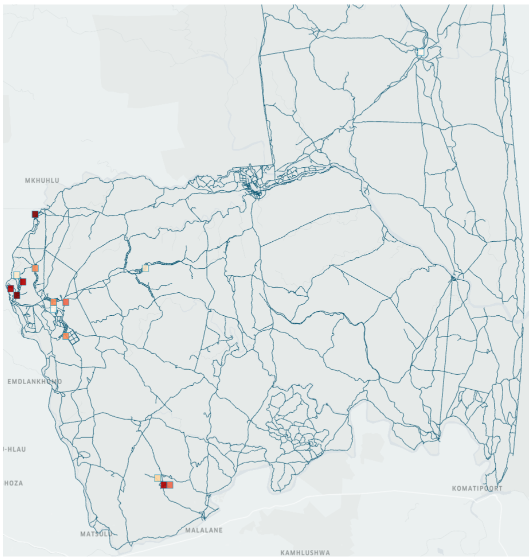

Figure 17: S1C1: Wildlife sightings locations and durations.

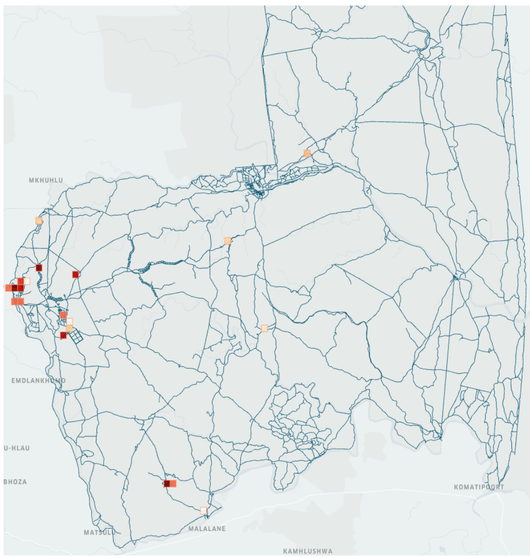

Figure 18: S1C2: Wildlife sightings locations and durations.

5.2. Scenario 2

In the following figures, the `TrafficGrid` layers (Figure 19 and Figure 20, respectively), the `TrafficJamGrid` layers (Figure 21 and Figure 22, respectively), and the `SightingsGrid` layers (Figure 23 and Figure 24, respectively) that resulted from S2C1 and S2C2 are presented side by side for comparison. Again, the darker the colour of a square, the more activity (i.e., traffic density, traffic jam duration, or wildlife sighting duration) occurred on the underlying road segment.

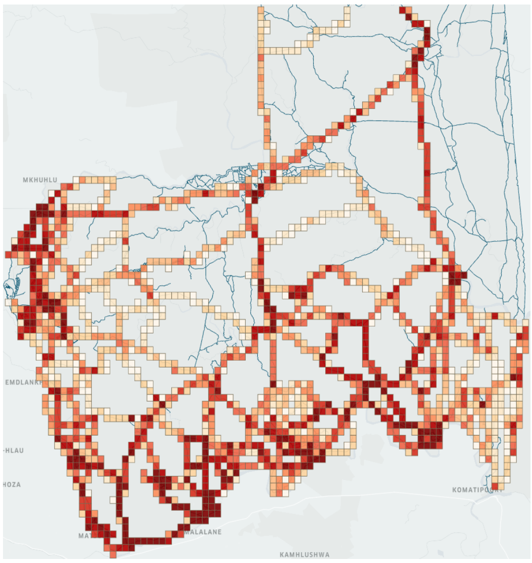

Figure 19: S2C1: Traffic density.

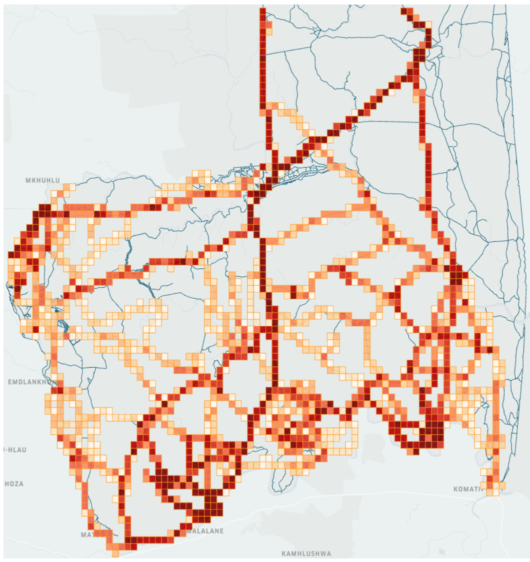

Figure 20: S2C2: Traffic density.

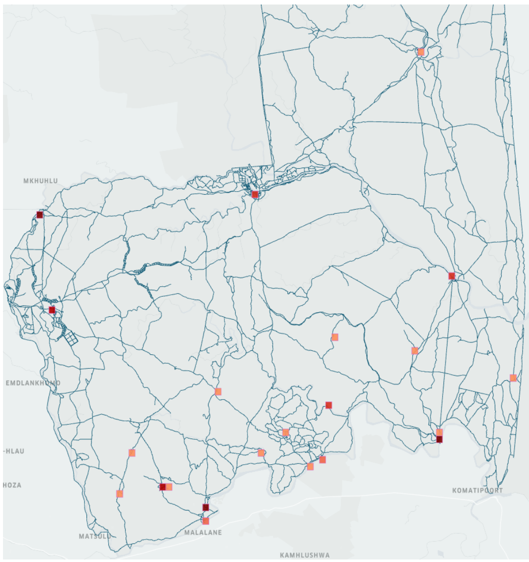

Figure 21: S2C1: Traffic jam locations and densities.

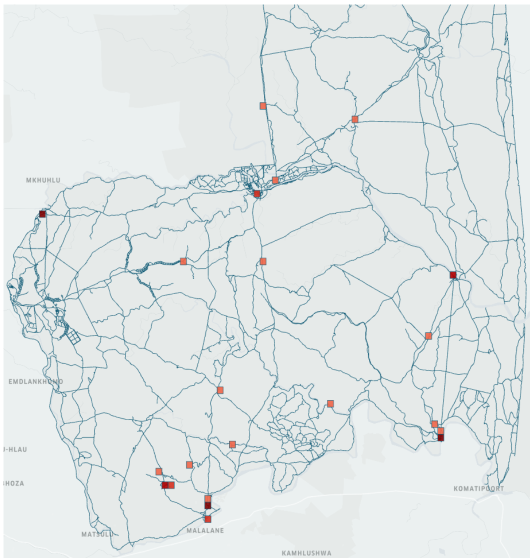

Figure 22: S2C2: Traffic jam locations and densities.

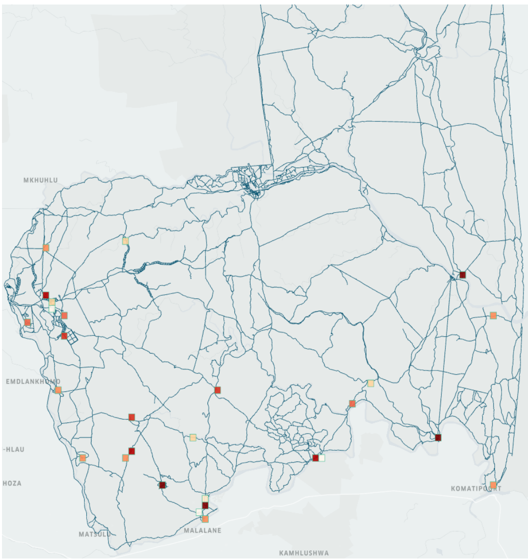

Figure 23: S2C1: Wildlife sightings locations and durations.

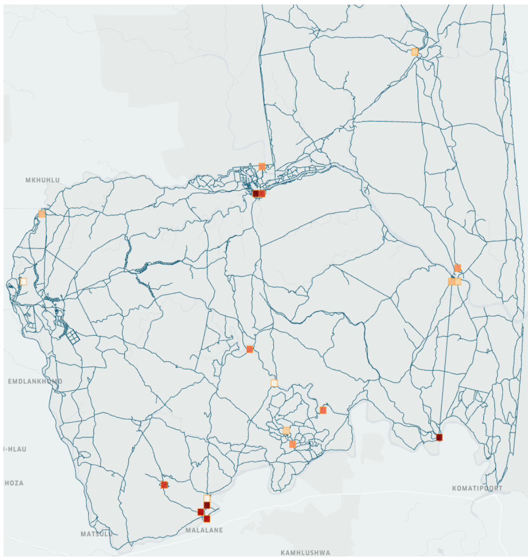

Figure 24: S2C2: Wildlife sightings locations and durations.

5.3. Travel Route Samples

To give a qualitative impression of how agents travel through the KNP in the model, two randomly sampled trajectories of `Visitor` agents are presented below. In Figure 25 and Figure 26, a roundtrip between Crocodile Bridge Gate and Skukuza Rest Camp is shown. In Figure 27 and Figure 28, a trip from Phabeni Gate to Paul Kruger Gate via Satara Rest Camp is shown. All travel trajectories include an exploratory phase, followed by a trip to the current destination via the shortest path. For an overview of the daily behaviour routine of the agents, see Section 3.1.4.

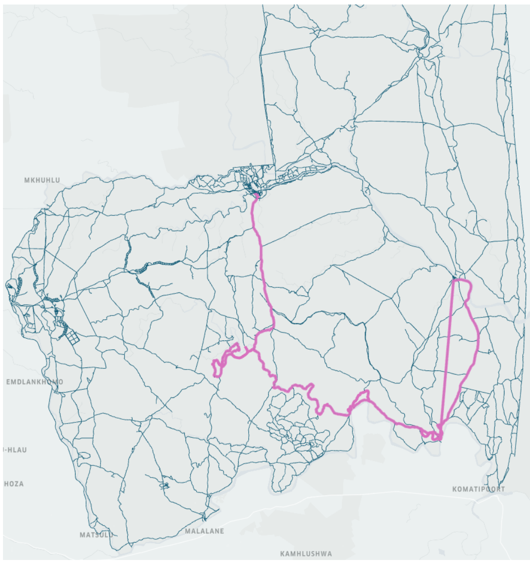

Figure 25: Outward route of a `Visitor` agent from Crocodile Bridge Gate to Skukuza Rest Camp.

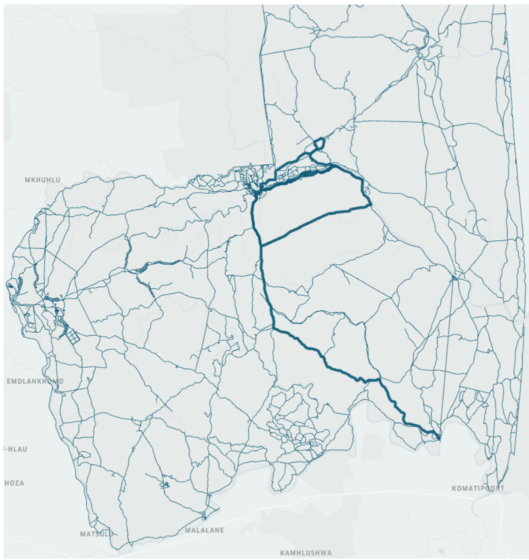

Figure 26: Return route of a `Visitor` agent from Skukuza Rest Camp to Crocodile Bridge Gate.

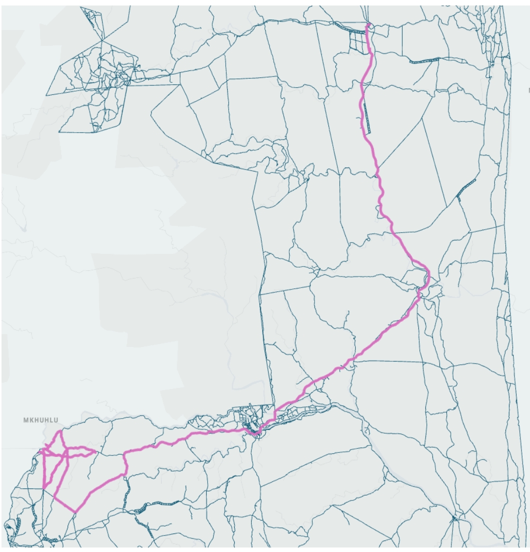

Figure 27: Outward route of a `Visitor` agent from Phabeni Gate to Satara Rest Camp.

Figure 28: Return route of a `Visitor` agent from Satara Rest Camp to Paul Kruger Gate.

6. Discussion and Evaluation

Based on the model description provided in Section 3.1 and the scenarios and results presented in Section 4 and Section 5, respectively, this section offers a critical discussion and evaluation of the results and the current development state of the model. The potential value and utility of the model for stakeholders is highlighted, and a list of features that can be improved or added to the model is provided.

6.1. Result Discussion

As mentioned in the disclaimer in the preamble of Section 4, the goal of the scenarios and result presentation was mainly to showcase the current functionality and potential value of the model. When viewed in isolation (i.e., outside the scope of this project) and based purely on scientific merit, the results presented in Section 5 are only moderately informative. The model tracks more information about the course of its simulations than was presented in the results of this report. However, the MARS Framework does not offer standardized tools for presenting and analysing this information meaningfully. This is because the focus of the MARS Framework is on modelling and simulation rather than on the analysis and presentation of simulation results. Therefore, many post-simulation analytical and visual functions that are specific to a given model need to be implemented on a case-by-case basis. Since the focus of this project was on designing and developing an ABM that models traffic flow in the KNP, such functions have been implemented only in rudimentary form during the course of the project.

Nevertheless, the results do show that traffic tends to cluster around entry points (i.e., the locations at which agents enter the KNP road network). In other words, agents tend to not quickly disperse from the POI from which they enter the KNP road network. Instead, they linger on the road segments in the immediate surroundings and only gradually spread out. This can be observed in, for example, Figure 13 and Figure 14. Both of these heatmaps contain dark red clusters around Pretoriuskop rest camp and Berg-en-Dal rest camp, which were specified as spawn locations in the scheduler CSV files (see Table 17 and Table 19). It has not been validated with domain experts whether this kind of behaviour is observed in the real-world KNP as well. It might be that the agents spend an above-average length of time in the relatively dense road network around the rest camps, causing the heatmaps to show spikes in those regions. If this is atypical behaviour relative to real-world observations, the behaviour of the agents might need to be refined accordingly.

As for traffic jam locations and wildlife sightings, the model appears to show a spatial correlation between the two. This can be observed when comparing, for example, Figure 15 and Figure 17. Many—but not all—locations at which traffic jams occurred are also locations at which wildlife sightings occurred. Another observation that reveals potential room for improvement in the movement logic of the agents is that many traffic jams and wildlife sightings happen in the vicinity of entry points. This might be coincidental; or it might be due to the fact that, based on the corresponding `TrafficGrid` heatmaps, these are also the regions in which agents spent the most time during the simulations.

In summary, the presentation of results has qualitative value, but requires improvement. It is especially important to add quantitative results to provide a more in-depth and specific view into the course simulations of the model.

6.2. Model Evaluation

In Section 6.1, it was mentioned that the result presentation requires improvement. When viewing the model as an end-to-end solution with three major requirements—including (i) a highly configurable simulation setup process that allows for a wide range of scenarios, (ii) an elaborate and detailed internal model structure that captures many of the intricacies of its real-world counterpart, and (iii) a model-specific process of result analysis and visualisation—then the third requirement would lower the overall quality of the model presented in this report. However, when considering only the first two requirements, the model offers a range of functions in terms of configurability and functionality that are relevant and valuable to its potential end users. These functions are not finished and polished; but as a prototype, they should be capable of showing the potential of such a model for providing decision support in questions pertaining to traffic and visitor management.

What might the application of this model by its end users look like? To answer this question, it is important to be mindful of the reduction property of models: a model never captures all aspects of its original, but only those aspects that are deemed relevant by the model creators and that *can* be captured at all (Stachowiak 1973). This is especially true for models of complex and potentially adaptive systems like traffic networks. Simulation models should not be viewed as a source of answers or truths, but rather as a source of possibilities and eventualities. A model that adequately captures the main attributes of its original can provide insights into the workings of a system that might not be readily attainable when studying the system itself. As such, the KNP visitor and traffic management model might serve as a platform for experimentation in a simulative setting. It can shed light on the impact of potential interventions that are too costly or otherwise too uncertain to execute without prior consideration.

6.3. Refinement and Addition of Features

As highlighted in parts of this report, model features have been implemented to varying degrees of completion. In addition, there are features that have not yet been implemented and that would add value to the model. In Table 6, a list of existing features that need to be refined and new features that need to be added to the model is presented. The list of based on the most recent evaluation of the current state of the model, but is not necessarily comprehensive.

Table 6: Refinement of existing features and addition of new features in potential future model versions.

Type	Number	Name	Description
Refinement	R1	Parametrisation of agents	Agent parameters should be extended (e.g., exploration duration, road surface preference, etc.).
	R2	Behaviour routines	Agent behaviour routines should be able to reliably cover a full day and be extensible to multiple days.
	R3	Driving speed	Agents should be able to drive below the speed limit during exploration.
	R4	Capacities of rest camps	Rest camp capacities should be integrated into the model for overnight agents to take into account.
	R5	Wildlife event spawn rates	Depending on the KNP region and the time of day, wildlife events should be spawned at different rates.
Addition	A1	KNP hours	Opening hours of the KNP should be integrated into the model for agents to take into account.
	A2	Road surfaces	Agents should consider road surface types during trip planning and travel.
	A3	Road access	Agents should obey road access restrictions during trip planning and travel.
	A4	Wildlife event types	Wildlife events should be of different types (e.g., common, normal, rare, special).
	A5	Simulation results	Results should feature more metrics and diagrams for a more comprehensive simulation summary.

7. Conclusion

In this project report, the prototypical model that was designed and developed during the course of the project was described. This description includes (i) an in-depth presentation of the current state of functionality of the model, (ii) an overview of the real-world datasets of the KNP that were provided by SANParks and that are integrated into the model and/or are available to inform configurations of the model, and (iii) a description of how the model currently can be configured by its potential end users. To assess the functionality of the model, some simulation scenarios were set up and run. These scenarios are hypothetical, but are based on insights gained during conversation with SANParks representatives. Qualitative simulation results were presented and critically evaluated.

In future developments and iterations of the model, current features might need to be tested more comprehensively and refined to capture more nuance of the underlying real-world traffic dynamics in the KNP. The list of features provided in Table 6 can serve as a starting point in such an endeavour, but is not necessarily comprehensive. Furthermore, an ongoing channel of communication and feedback loop with domain experts would continue to ensure that the model remains as close and true to its original as possible. Lastly, as indicated during the discussion and evaluation of this report, the presentation of results in a manner that can help stakeholders make more informed decisions on interventions in visitor and traffic management would be the greatest improvement to the current state of the project.

Glossary

ABM An agent-based model (ABM) is a computerised simulation model in which a set of agents share a common environment. 1, 2

Agent An autonomous software entity that is situated in a virtual environment and executes behaviour to achieve its goals over time. 2

Configuration A specification of the initial parameter values of a simulation of an ABM. 2

Environment The spatial extent of an ABM in which agents are situated and can move. 2

Layer A cross-section of the environment of a MARS model that contains like objects. See Section 2.2 for more details. 3

MARS Framework The Multi-Agent Research and Simulation (MARS) Framework is a programming framework that enables the implementation and simulation of ABMs. See Section 2.2 for more details. 1

POI A point of interest (POI) is a georeferenced location in the environment of a MARS model that agents can navigate to to satisfy some need. In the model, a POI is described by a position, name, type, and other information. See Appendix A for a full listing of POI types, and Appendix B for some POIs with public access. 5

Scheduler A component of a MARS model that enables the addition of new agents to an ongoing simulation. See Section 3.1.6 and 3.3.2 for more details. 9

Simulation An execution of consecutive iterations of the interaction cycle between agents and the environment of an ABM over some time period (see Figure 2). 1

Simulation step A single time step (i.e., the unit of time) in the simulation of an ABM. 2

Spawn event An event during a spawn period in which new agents join a simulation. See Section 3.1.6 and 3.3.2 for more details. 9

Spawn period A time period in a simulation of a MARS model during which spawn events occur. See Section 3.1.6 and 3.3.2 for more details. 9

UTC Coordinated Universal Time (UTC) is a global standard for measuring time. It has defined a set of standardised formats for expressing dates and time. The MARS Framework uses some of these formats to specify temporal features of a simulation or a spawn period (such as its start time and end time). 10

WKT Well-Known Text (WKT) is a standardised text format for the representation of vector geometry objects. It is used in the MARS Framework to define and describe georeferenced point, line, and polygon features in the environment of a model. 15

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A. POI Types and Counts

In Table 7, the types of POIs that are integrated into the model are listed. These can be used for `sourceType` and `targetType` in scheduler CSV files (see Section 3.1.6 and 3.3.2 for more details). The underlying dataset was provided by SANParks (see Section 3.2.1 for more details).

Table 7: All types of POIs that are available in the model, including counts and access restrictions.

Type	Count	Access
Bird hide	13	Public
Borehole	76	
Bushveld camp	5	
Dam	5	
Dam, concrete	12	
Dam, concrete weir	7	
Dam, earthen	19	
Day visitor	8	
Get out point	13	
KNP Gate	10	
Look out point	13	
Pan	1	
Picnic spot	14	
Pipeline trough	6	
Port of Entry	2	
Rest camp	12	
Satellite camp	5	
Total Public	**221**	
4X4 Camp	1	Private
Backpack camp	1	
Bush Lodge	3	
Concession camp	29	
EE Centre	1	
HR camp	2	
Private Gate	1	
SANDF Base	5	
Trails camp	8	
Unspecified	3	
Total Private	**54**	
Kennels/ K9 training	1	Staff
Ranger camp	11	
Staff Accommodation	11	
Total Staff	**23**	
Grand Total	**298**	

B. POIs with Public Access

In the following tables, some publicly accessible POIs that are integrated into the model are listed. These can be used for `sourceName` and `targetName` in scheduler CSV files (see Section 3.1.6 and 3.3.2 for more details). The underlying dataset was provided by SANParks (see Section 3.2.1 for more details).

Table 8: KNP Gate and Port of Entry POIs.

Type	Name
KNP Gate	Crocodile Bridge
	Malelane
	Numbi
	Orpen
	Orpen Boom
	Pafuri
	Paul Kruger
	Phabeni
	Phalaborwa
	Punda Maria
Port of Entry	Giriyondo Port of Entry
	Pafuri Port of Entry

Table 9: Rest camp POIs.

Type	Name
Rest camp	Berg-en-Dal
	Crocodile Bridge
	Letaba
	Lower Sabie
	Mopani
	Olifants
	Orpen
	Pretoriuskop
	Punda Maria
	Satara
	Skukuza
	Shingwedzi

Table 10: Get out point POIs.

Type	Name
Get out point	Albasini Ruins
	Kruger Tablets
	Longwe
	Look out
	Mathekenyane - Granokop
	N’wamanzi
	Nkumbe
	Olifants
	Orpen Dam
	Rabelais’ Hut
	Red Rocks
	Stevenson Hamilton
	Tshanga

Table 11: Look out point POIs.

Type	Name
Look out point	Baobab Tree
	Bobbejaankrans
	Dzundwini
	Hippo Pool
	Matjulu
	Mingerhout
	Mondzweni
	Nyamundwa
	Olifant Drinkgat
	Renosterpan
	Silwervis
	Thulamela
	View point

Table 12: Bushveld camp POIs.

Type	Name
Bushveld camp	Bateleur Biyamiti Shimuwini Sirheni Talamati

Table 13: Satellite camp POIs.

Type	Name
Satellite camp	Balule Malelane Maroela Tamboti Tsendze

Table 14: Bird hide POIs.

Type	Name
Bird hide	Biyamiti Bird Hide Coetzee Dam Bird Hide Gardenia Bird Hide Kanniedood Bird Hide Lake Panic Bird Hide Matambeni Bird Hide Nthandanyathi Bird Hide Nyawutsi Bird Hide Pioneer Dam Bird Hide Ratel Pan Bird Hide Sable Dam Bird Hide Shipandani Bird Hide Sweni Bird Hide

Table 15: Picnic spot POIs.

Type	Name
Picnic spot	Afsaal Babalala Makhadzi Masorini Mlondozi Dam Mooiplaas Muzandzeni Nhlanguleni Nkuhlu Nwanetsi Pafuri Punda Maria EE Centre Timbavati Tshokwane

C. Configurations for Simulation Scenarios

In this section of the appendix, the scheduler configurations for Scenario 1 (S1) and Scenario 2 (S2) are listed. For an overview of the scenarios, see Section 4.

Scenario 1: Scheduler Configurations

In the following four tables, the scheduler configurations for Configuration 1 (S1C1, Table 16 and Table 17) and Configuration 2 (S1C2, Table 18 and Table 19) are listed.

Table 16: S1C1: Configuration of `VisitorScheduler` parameters.

Spawn period	startTime	endTime	spawning-IntervalInMinutes	spawning-Amount	sourceName	sourceType	targetName	targetType
1	6:00	7:00	30	7	Phabeni	KNP Gate		
2	6:45	9:45	20	4	Numbi	KNP Gate		

Table 17: S1C1: Configuration of `OsvTourGuideScheduler` parameters.

Spawn period	startTime	endTime	spawning-IntervalInMinutes	spawning-Amount	sourceName	sourceType	targetName	targetType
1	6:00	6:15	15	3	Berg-en-Dal	Rest camp		
2	6:00	6:15	15	3	Pretoriuskop	Rest camp		
3	9:00	9:15	15	3	Berg-en-Dal	Rest camp		
4	9:00	9:15	15	3	Pretoriuskop	Rest camp		

Table 18: S1C2: Configuration of `VisitorScheduler` parameters.

Spawn period	startTime	endTime	spawning-Interval-InMinutes	spawning-Amount	sourceName	sourceType	targetName	targetType
1	6:00	7:00	30	3	Numbi	KNP Gate		
2	7:00	8:00	30	2	Phabeni	KNP Gate		
3	8:00	9:00	30	3	Numbi	KNP Gate		

Table 19: S1C2: Configuration of `OsvTourGuideScheduler` parameters.

Spawn period	startTime	endTime	spawning-Interval-InMinutes	spawning-Amount	sourceName	sourceType	targetName	targetType
1	6:00	6:30	15	2	Berg-en-Dal	Rest camp		
2	6:00	6:30	15	2	Pretoriuskop	Rest camp		
3	7:00	8:00	30	3	Phabeni	KNP Gate		
4	9:00	9:30	15	2	Berg-en-Dal	Rest camp		
5	9:00	9:30	15	2	Pretoriuskop	Rest camp		
6	9:00	9:30	30	3	Numbi	KNP Gate		

Scenario 2: Scheduler Configurations

In the following two tables, the scheduler configurations for Configuration 1 (S2C1, Table 20) and Configuration 2 (S2C2, Table 21) are listed.

Table 20: S1C2: Configuration of `VisitorScheduler` parameters.

Spawn period	`startTime`	`endTime`	`spawning-Interval-InMinutes`	`spawning-Amount`	`sourceName`	`sourceType`	`targetName`	`targetType`
1	6:00	7:00	15	10	Crocodile Bridge	KNP Gate		
2	6:00	7:00	20	10	Malelane	KNP Gate		
3	6:00	7:00	20	10	Phabeni	KNP Gate		
4	7:00	8:00	15	10	Crocodile Bridge	KNP Gate		
5	7:00	8:00	20	10	Malelane	KNP Gate		
6	7:00	8:00	20	10	Phabeni	KNP Gate		
7	8:00	9:00	20	10	Crocodile Bridge	KNP Gate		
8	8:00	9:00	30	10	Malelane	KNP Gate		
9	8:00	9:00	30	10	Phabeni	KNP Gate		

Table 21: S2C2: Configuration of `VisitorScheduler` parameters.

Spawn period	`startTime`	`endTime`	`spawning-Interval-InMinutes`	`spawning-Amount`	`sourceName`	`sourceType`	`targetName`	`targetType`
1	6:00	7:00	30	10	Crocodile Bridge	KNP Gate		
2	6:00	7:00	40	10	Malelane	KNP Gate		
3	6:00	7:00	40	10	Phabeni	KNP Gate		
4	7:00	8:00	30	10	Crocodile Bridge	KNP Gate		
5	7:00	8:00	40	10	Malelane	KNP Gate		
6	7:00	8:00	40	10	Phabeni	KNP Gate		
7	8:00	9:00	40	10	Crocodile Bridge	KNP Gate		
8	8:00	9:00	60	10	Malelane	KNP Gate		
9	8:00	9:00	60	10	Phabeni	KNP Gate		